[Thesis Check-in] Dynamic Networks and Emotional Catalysts: A Computational Analysis of Mobilization in Digital Activism

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Research Background

Background:

- Social media platforms like Twitter and Reddit have transformed social mobilization, enabling rapid information diffusion and decentralized coordination (e.g., Arab Spring, Black Lives Matter).
- While outcomes are visible, the dynamic mechanisms of how high-intensity emotional events reshape social networks and drive large-scale mobilization remain unclear.

Key Problem:

- How do emotional events dynamically activate dormant ties, form new connections, and reorganize network structures to enable rapid diffusion?

Research Gap and Contribution

Research Gap:

- *Cause of Gap:* Methodological limitations in prior research:
  - **Qualitative Approaches:**
    - Example: Tufekci's *Twitter and Tear Gas*.
    - Relied on interviews and observations but lacked tools to capture real-time network evolution.
  - **Static Snapshot Comparisons:**
    - Example: Facebook emotional contagion experiment.
    - Compared static pre- and post-intervention networks, missing continuous changes and feedback loops.

Contribution:

- Address methodological gaps by analyzing dynamic network changes, capturing continuous changes and interactions during high-emotion events.

Research Question

Central Question:

- How do high-intensity emotional events dynamically alter social network structures, and how do these changes influence information diffusion and collective mobilization?

To anwser:

- How do emotional events activate dormant ties or create new connections?
- What role do these structural changes play in reducing barriers to information diffusion?
- How do these processes contribute to reaching critical mass and sustaining collective action?

Key Concepts and Relationships

Independent Variable: High-Intensity Emotional Events

- Examples: Death of George Floyd, major political crises.
- Impact:
  - Emotional contagion spreads heightened emotions like outrage or solidarity.
  - Catalyzes structural changes by activating dormant ties, forming new connections, and reinforcing existing links.

Dependent Variables:

- **Dynamic Structural Changes:**
  - Activation of weak ties and bridging nodes.
  - Reduction of resistance to information flow.
- **Information Diffusion and Mobilization:**
  - Speed and scale of diffusion.
  - Reaching critical mass—tipping point for sustained collective action.

Data Collection Plan

Data Source:

- Using the scraper developed by Apify platformfor X (formerly Twitter) due to its affordability (1/100 of Twitter API cost).
- Focus on posts related to the Black Lives Matter movement during its peak period.

Collection Period:

- Start: April 2020.
- Duration: One year.

Baseline Dataset:

- At least 100,000 posts.
- Metadata: Author, timestamp, retweet relationships, emotional content.

Data Processing Plan

Network Construction:

- Nodes: Users.
- Edges: Retweet relationships.
- Output: Daily social network graphs (at least 365 snapshots).

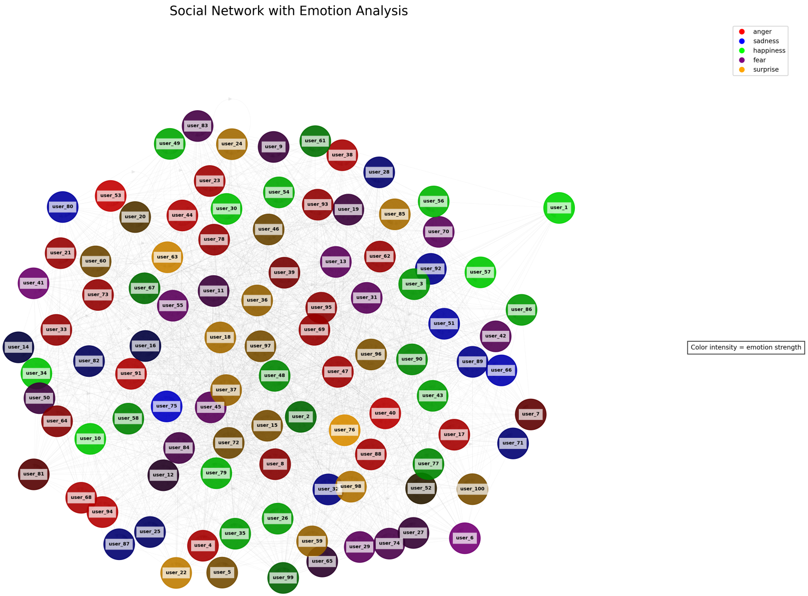
Emotion Annotation:

- Tools: BERT or ChatGPT API (final choice based on testing).
- Annotation Target: Emotional intensity and direction of posts.

Sample data in text


```json
[
    {
        "post_id": 1,
        "author": "user_96",
        "timestamp": "2021-08-28T07:49:03.514192",
        "content": "YO!! 😱 cops just joined our protest?! took a knee w/ us n everything!!/ never thought id see this fr fr ✊🏾 #BLM #WeOutHere",
        "emotion": "surprise",
        "emotion_intensity": "High",
        "retweets": [51, 68, 66, 30, 32, 98, 21, 94, 40, 41, 86, 43, 25]
        },
        {
        "post_id": 2,
        "author": "user_92",
        "timestamp": "2021-03-20T13:13:15.857729",
        "content": "watching this trial rn... can't stop crying 💔 when will it end?? system been broken for too long smh #JusticeForGeorge #blm",
        "emotion": "sadness",
        "emotion_intensity": "medium",
        "retweets": [45, 3, 64, 90, 63, 19, 44, 24, 48, 83, 1]
        },
```

Sample data in graph



Methods Overview

Step 1: Sentiment Analysis

- Use pre-trained models (e.g., BERT or ChatGPT API) to classify emotions in tweets.
- Identify the dominant emotion (e.g., sadness, anger, joy) and its intensity (low, medium, high).
- Purpose: Capture emotional triggers driving network interactions and diffusion.

Step 2: Social Network Construction

- Represent users as nodes and retweet relationships as edges.
- Create daily network graphs to track temporal and structural changes.
- Output: A sequence of network snapshots for dynamic analysis.

Methods Overview

Step 3: Dynamic Graph Neural Networks

- Integrate emotion labels as node features to model dynamic changes.
- Simulate how emotions influence structural evolution, such as activating latent ties or forming new connections.
- Predict key events like achieving critical mass or accelerating diffusion.

Step 4: Model Validation

- Compare predictions with observed patterns to validate accuracy (e.g., bridging nodes, weak ties activation).
- Test model generalizability using unseen datasets from other social movements (e.g., Arab Spring, Women's March).
- Ensure robustness across diverse social contexts and dynamics.

Expected Results and Research Questions

Expected Results:

- **Structural Changes:** High-intensity emotional events activate latent ties, create new connections, and reinforce existing relationships.
  - Emergence of bridging nodes reduces resistance to information flow.
- **Critical Mass:** Structural shifts accelerate information diffusion, allowing the network to reach the tipping point for sustained collective action.
- **Model Generalization:** The Dynamic Graph Neural Network model identifies similar patterns in unseen datasets, validating the mechanisms across different social movements (e.g., Arab Spring, Metoo).

Connection to Research Questions:

- Explains how emotional events dynamically reshape network structures.
- Demonstrates how these structural changes drive large-scale mobilization through critical mass.
- Confirms that the findings are applicable across diverse movements.

Main Challenges and Solutions

Challenges:

- **Efficient Scraping Logic:**
  - Scaling from single-day to year-long data requires avoiding redundant scraping.
  - A well-designed logic is critical to saving time and reducing API costs.
- **Dynamic Graph Neural Networks:**
  - Applying GNNs to study social networks is a new approach, proposed only in the past two years.
  - Limited prior research means fewer methodologies to reference.

Planned Solutions:

- Leverage deep learning and explainable machine learning courses to refine methods.
- Iteratively test and develop the scraping logic for long-term data collection.

Thank You and Question section